



THE ADVENTURES OF DR. CIRCUIT · FAMILY STORY

# HYDRO ELECTRIC

*The wheel, the creek, and the sunshine in disguise*

Starring Natalia, Dallas, Bro — and Zippy the Zap

## BEFORE THE SUN

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Saturday started before the sun did.

Her dad, **Dallas**, woke **Natalia** the way he always did on fishing days – not with words, but with the smell of coffee and the clatter of gear by the front door. Two rods. One tackle box. A thermos, two sandwiches wrapped in wax paper, and a jug of water already sweating through the plastic.

Natalia was dressed in eleven seconds flat. Fishing was her favorite thing in the world, but that wasn't quite the whole truth. Fishing was her favorite thing to do *with her dad* – and that had been true for as long as she could remember.

**Bro** was already waiting at the truck. White as a cloud, short-coated, with a back as broad as a small pony. Everyone who saw him asked if he was a Great Pyrenees, and Dad always said, "Close. He's on loan from the polar bears." Bro took up the entire truck bed and saw no problem with this.

Dallas turned the key, then turned it back off. "Gotta tie the rods down first," he said. "No sense burning gas just to sit in the driveway."



## PART TWO

# THE OLD COOPER PLACE

The creek ran behind the old Cooper place, twenty minutes out of town, past two cattle gates and a bridge that rattled like a bucket of bolts. The fishing hole was a slow green bend shaded by big old sycamores, and the water moved through it the way it had for a hundred years – patient, unhurried, going wherever it was going.

They set up on the bank. Bro waded in to his chest, stood perfectly still like he was thinking about something, then shook himself so hard the whole creek got a second rain.



"Bro," Natalia said. "You're scaring the fish."

"He's not scaring them. He's *announcing* them," Dallas said. "Politely."



Natalia cast. The bobber landed soft and drifted. Dragonflies stitched the air.

Down the bank, past a stand of cattails, the old mill sat half in the water — gray boards, a tin roof gone green and gray, and on its side, a wheel. Big as a barn door, half its paddles furred with moss, turning. Slow. Steady. Never once in a hurry.

"Dad," Natalia said, watching it instead of her bobber. "What's that for?"



## PART THREE

# THE PUSH

Dallas reeled in a little slack and sat back on the tackle box.

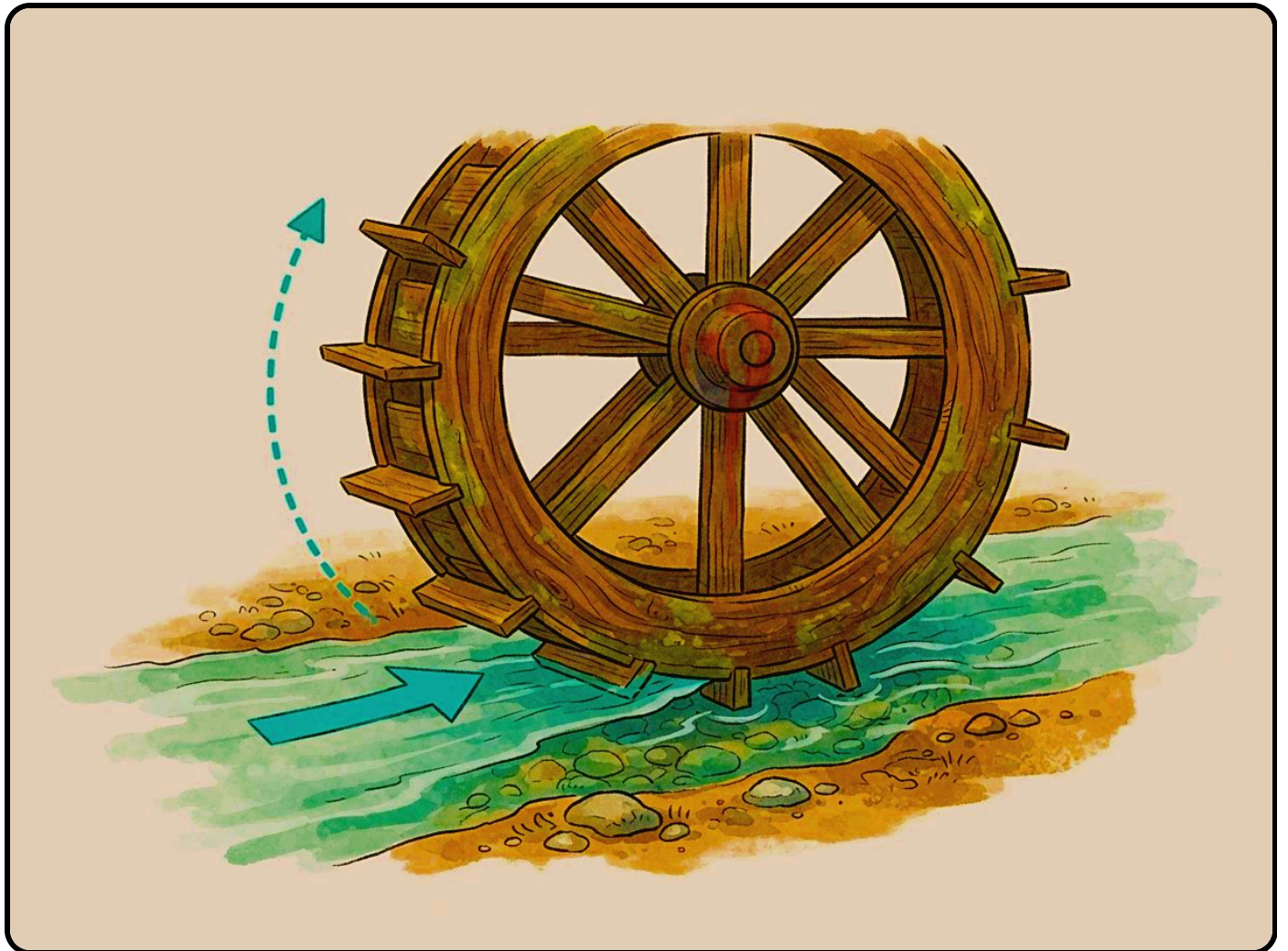
"That's a **water wheel**. Old Mr. Cooper's corn mill used to run off it — ground corn into meal for half the county. A neighbor bolted a little generator in there a few summers back, so the old wheel still has a job. But look at what it's doing right now."

"It's turning."

"It's turning," he agreed. "Why?"

"The water's pushing it."

"Right. Water comes downstream, hits a paddle, pushes. That push is energy — moving water is one of the oldest kinds of power there is. And the wheel doesn't work alone. See that axle going into the mill? The wheel turns the axle. The axle turns gears — and the gears speed the turning up. That fast spin turns a shaft that drives a machine called a **generator**."



"What does a generator do?"

"It takes turning — plain old spinning — and changes it into electricity. It's got magnets inside, and coils of wire. Spin those magnets past the wire fast enough, and they nudge the tiny bits inside it along. Those bits are **electrons**. The push runs down the wires, out of the mill, down the road —"

"To houses?"

"To houses. Lights. Fridge. Maybe the fan in your window you can't sleep without."



Natalia squinted at the wheel. "So the water does all that."

"The water does all that. Nobody has to feed it or pay it. It comes downhill like it was going to anyway, and we borrow that push on the way past."

"So it's like my bike light," she said. "The little roller thing on the tire. The wheel turns it and the light comes on."

Dallas grinned. "Same idea. Exactly the same idea. Except the wheel is about a million times bigger, and the creek never gets tired."

She watched the water slap a paddle. The paddle came up dripping, and the wheel went around anyway – slow and sure, like it had all the time in the world. Which it did.

"Energy doesn't disappear," Dallas said. "It just changes clothes. Water's moving – that's one outfit. Wheel's turning – new outfit. Electricity in the wires – new outfit again. Same energy the whole time. People call this kind **hydro electric**. **Hydro** is just an old word for water."

## PART FOUR

# ZIPPY

Then something on the wheel moved.

Not a paddle. Not a leaf.

A small yellow figure was sitting on the crossbeam, listening – bolt-shaped, goggle-topped, a little zigzag of sparks crackling off one hand like a firefly in a jar.

**"ZAP!"** it said, and bounced to its feet. "Sorry. Sorry! I was going to wait, I really was, but you got it *exactly* right, and I couldn't sit still another second."



Natalia's bobber drifted, forgotten. "Who *are* you?"

**"Zippy.** Zippy the Zap. Professional spark." He hopped down to the mill rail, and the air smelled like a just-struck match and summer rain. He pointed with his whole hand. "And your dad – he even got the magnets right. Most people forget the magnets."

"Most people?" said Dallas.

"Most *everybody*." Zippy crackled. "Magnets spin past wire, wire pushes its electrons, the push goes down the line. No spin, no zap – that's the rule for generators. **Motion plus magnets** – most generators on Earth, from this old wheel to the biggest dam on the creek, play by it."

"Motion plus magnets," Natalia said, testing it.



"Motion plus magnets," Zippy said, and the wheel slapped the water, and the sparks around him crackled like they agreed.

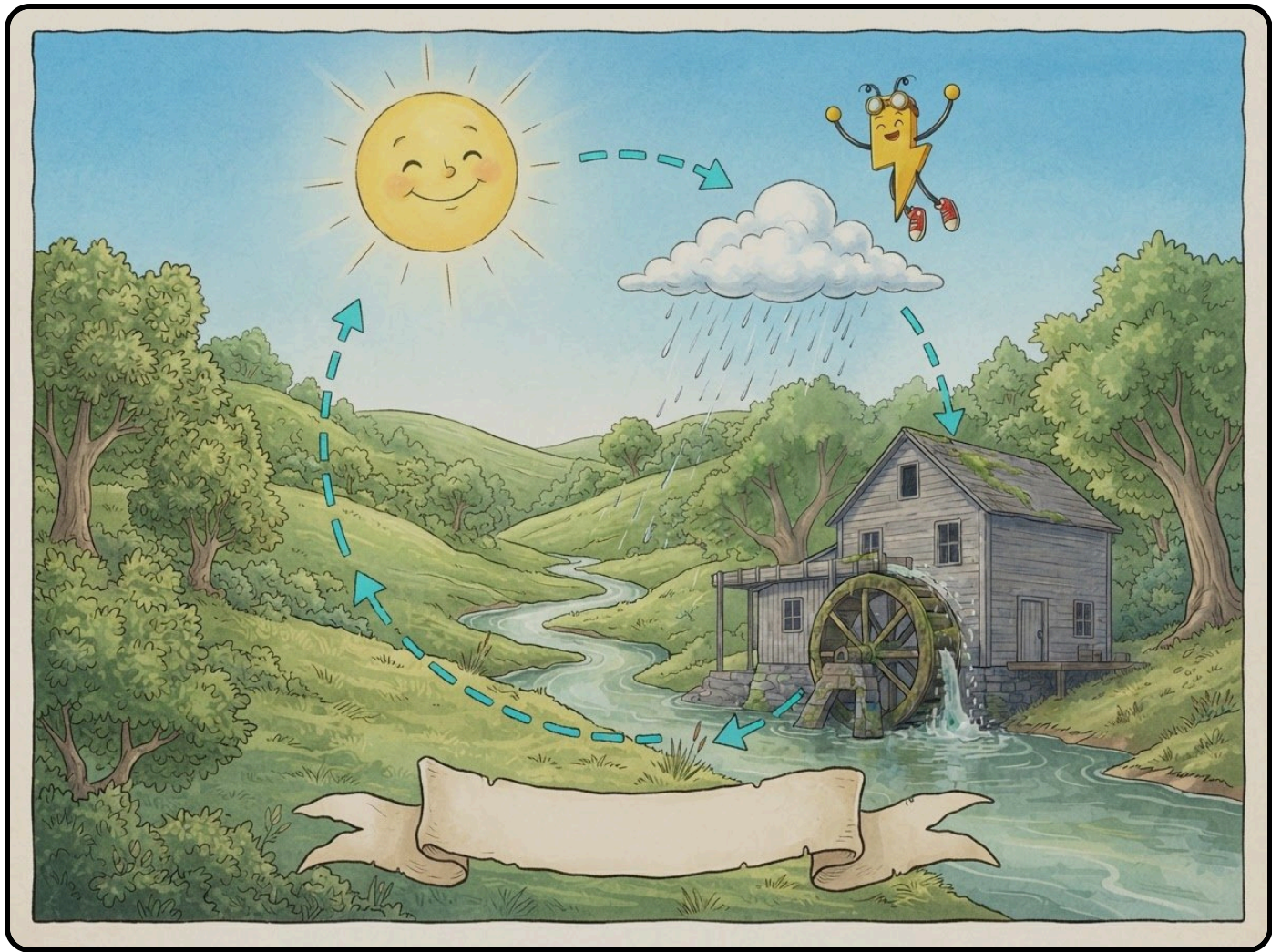
Then he went very still. Which, for Zippy, lasted most of a second.

"But where do you think the water got its push?"

"Um. The rain?"

"From the sky, yeah. But who put it *up* in the sky?" He flung both arms up at the sun. "That guy. The sun lifts most drops out of the ocean, tucks them into a cloud, and drops them on the hills. And now it's come all the way downhill for one more job." He grinned so wide it nearly split his face. "This old wheel runs on water, kid – and the water runs on **sunshine**, dressed up like a river."

Dallas laughed and shook his head. "Huh. Never thought of it that way."



"Nobody ever does. That's my favorite part." Zippy checked the sun, checked the wheel, checked the two of them. "All right. I've bothered you enough. But you—" He pointed with his whole hand. "You asked the right question. That's the job, you know."

"The job?"

"**Junior Hydro Engineer.**" He gave the mill rail a good hard stomp, and sparks popped off his sneakers in a bright little crown. "You're hired."

And then – **ZOOM**. He was gone up the creek in a yellow streak, leaving one zigzag of light hanging in the air like a name written in the sky.

Bro stared after him for a long moment. Then he huffed, thoroughly unimpressed, and went back to sniffing the tackle box.

## SET THE HOOK

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Her bobber went under.

"*Set the hook* – set it, set it!"

She set it. The rod bent and the line cut a bright little V through the green water, and after a fight that felt much longer than it was, she brought a smallmouth bass up onto the bank – thick, bronze-sided, gills opening and closing slow. Bro sniffed it once and sat down in approval.



"Look at him," Dallas said. "Good fish. You know this creek gave you two things today."

"One fish."

"One fish, and one turn of that wheel. Same water."

Natalia tucked a wet strand of black hair behind her ear and looked down the bank, where the wheel went around and around – the low sun turning his beard the same coppery orange as the rust on the mill's old hub.

They eased the bass back into the shallows. It hung there for a second, deciding, then shot off into the green dark.



Natalia didn't cast again right away. She just watched the wheel — and this time it looked different. Not like an old thing left over from a long time ago, but like something still working, still turning, still passing the day's push along to somebody's light.

"Dad?"

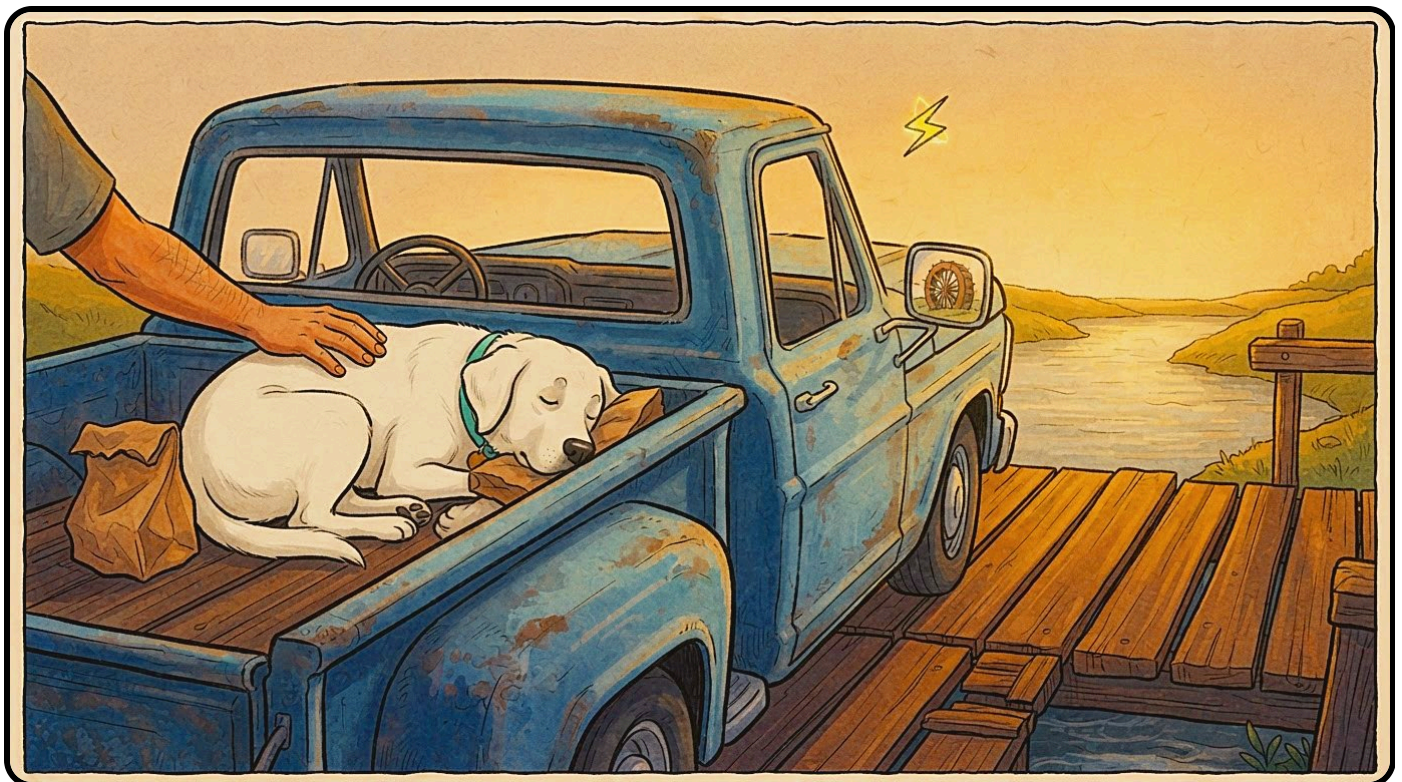
"Yeah."

"When I'm big, can we build one?"

Dallas capped the thermos and stood up. "I don't see why not. You've got the hard part figured out already."

"What's the hard part?"

"Knowing what you're looking at."



They loaded up. Bro claimed the whole truck bed, as usual, and put one enormous paw on the sandwich bag. The truck rattled over the bridge. In the rearview mirror the wheel kept going —

one paddle, then the next, then the next – the whole creek going round, and coming out as light.



Hydro Electric

A story for Natalia, from Dad